

Adriana Caraeni

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EDUCATION

University of Massachusetts Amherst

Sep. 2023 – May 2027

B.S. in Computer Science, Applied Mathematics | Business Minor | Commonwealth Honors College (GPA: 3.96)

Amherst, MA

- **Relevant Coursework:** Data Structures, Programming Methodology, Artificial Intelligence, Multivariate Calculus, Statistics, Object-Oriented Programming, Linear Algebra, Differential Equations, Discrete Math, Computer Systems Principles
- **Honors:** NCWIT Collegiate Award National Nominee, Goldwater Scholarship 2025 UMass Nominee, Chancellor's Award Scholarship (4X), Crowley-Nowick Scholarship (2X), Phi Kappa Phi Inductee, Hack(h)er Best Lottery Hack 2024

PUBLICATIONS

Evaluating GPT-4 at Grading Handwritten Solutions in Math Exams

LAK Conference 2025

HiPO: Hierarchical Preference Optimization for Adaptive Reasoning in Large Language Models

under review

TECHNICAL SKILLS

Languages & Frameworks: Python, Java, C, JavaScript, TypeScript, React.js, Node.js, HTML/CSS

Tools & Technologies: Git/GitHub, Vite, Docker, AWS, MongoDB, Firebase, Digital Ocean, REST APIs, Socket.IO, JUnit, Agile

Machine Learning: TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, HuggingFace, LLMs, RunPod

TECHNICAL EXPERIENCE

Commonwealth of Massachusetts

May – August 2025

Artificial Intelligence Intern

Amherst, MA

- Built an automated translation system that processes videos into multiple target languages concurrently by building off AWS video transcriber repository with parallel Google Translate API execution serving **more than 1,200** Culinary Career Ladder students.
- Achieved **95%** reduction in operational expenses by implementing intelligent transcription caching that reuses segments for duplicate content, deploying scalable compute infrastructure, and establishing automated processing pipelines.
- Enhanced translation workflow efficiency by **83%** through unified content management and seamless save/edit functionality.
- Created an email agent for the IT department utilizing UMass GenAI Playground, reducing time needed for drafting emails by **50%**.

Teamup, Apps for Good Nonprofit: CapMetro

June - August 2025

Team Lead / Full Stack Developer

Remote

- Led a 9-person development team to build a React.js web application displaying real-time CapMetro bus locations with **15-second refresh** intervals using WebSocket, integrating Texas Open Data Portal API and Google Maps for public transit accessibility.
- Architected scalable frontend using Vite and Tailwind CSS, led team implementation of real-time tracking features, and personally executed Digital Ocean deployment, developed WebSocket integration, and curated system design.
- **Technologies:** React.js, Vite, Tailwind CSS, Google Maps API, WebSocket, RESTful APIs, Digital Ocean

Minutemen Alternative Investment Fund

Feb 2025 – May 2025

ARQ-Fellow, Quantitative Researcher

Amherst, MA

- Developed and presented Python-based trading algorithm exploiting S&P 500 inclusion momentum, implementing MLP neural network model with 12 technical indicators to predict optimal exit timing within 1-7 days post-inclusion.

RESEARCH

Algoverse AI Research

June 2025 – Present

- Developed HiPO framework extending DPO with span-level auxiliary losses, implementing multi-objective training with PyTorch on RunPod GPU infrastructure for targeted optimization of reasoning components in 7B parameter models.
- Engineered end-to-end training pipeline with custom DataLoader, tokenization handling, and automated gradient computation across Meta-thinking, Tactical, and Answer spans using Transformers library and AdamW optimization.
- Designed comprehensive evaluation framework using LLM-as-a-Judge methodology, assessing model performance across coherence, accuracy, goal completion, and verbosity metrics beyond traditional token-level loss measurement.

Center for Intelligent Information Retrieval Lab

Sept 2024 – May 2025

- Developed a series of web pages to be deployed on Amazon Mechanical Turk in order to gather data on LLM preferences.
- Researched methods to enhance Large Language Models (LLMs) for improved personalization in user-specific question answering.
- Conducted mechanistic interpretability analysis on RankLlama 7B model by probing 10 retrieval-specific features (e.g., BM25, TF-IDF components) to identify correlations between activation patterns and traditional IR scoring formulas.

Machine Learning for Education Lab

May 2024 – Sept 2024

- Designed a multi-tiered batch prompting system for ChatGPT-4, improving grading accuracy by **20%** for real student exams.
- Developed and integrated data collection and evaluation pipelines for GPT-4o in an NLP research project, including implementing rubric-based prompting, chain-of-thought (CoT) generation, zero- and few-shot learning techniques, and aligning with standard data splits and metrics for reproducibility.

PROJECTS

Artemis: Integrated Marketing Automation Platform

- Architected full-stack email marketing platform with AI-powered drafting, multi-account management, campaign scheduling, and SMTP/IMAP integration for automated email sending/receiving.